

12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

Sheet: /
File: XG_Mobile_USB_Hub.kicad_sch

Title: XG Mobile USB Hub

Size: A3Date:KiCad E.D.A. 9.0.9

Rev: DRAFT AId: 1/4

USB–C DAUGHTERBOARD / ENGINEERING DRAFT

Upstream: USB–C, USB 3.2 Gen 2 data UFP, USB–PD power source, 20 V / 5 A maximum.
Downstream: 2 Type–C + 3 Type–A SuperSpeed, 1 Type–A USB 2.0.
Use two GPU power inputs OR one EPS input. Not for simultaneous independent supplies.

Power Supplies and Inputs

File: XG_Mobile_USB_Hub_Power.kicad_sch

MCU and Upstream PD Control

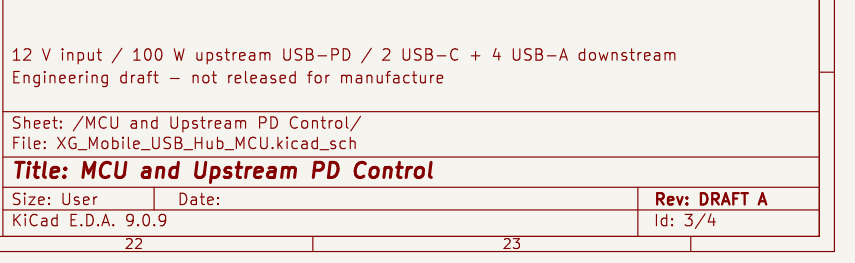
File: XG_Mobile_USB_Hub_MCU.kicad_sch

USB Hub and Downstream Ports

File: XG_Mobile_USB_Hub_USB_Hub.kicad_sch

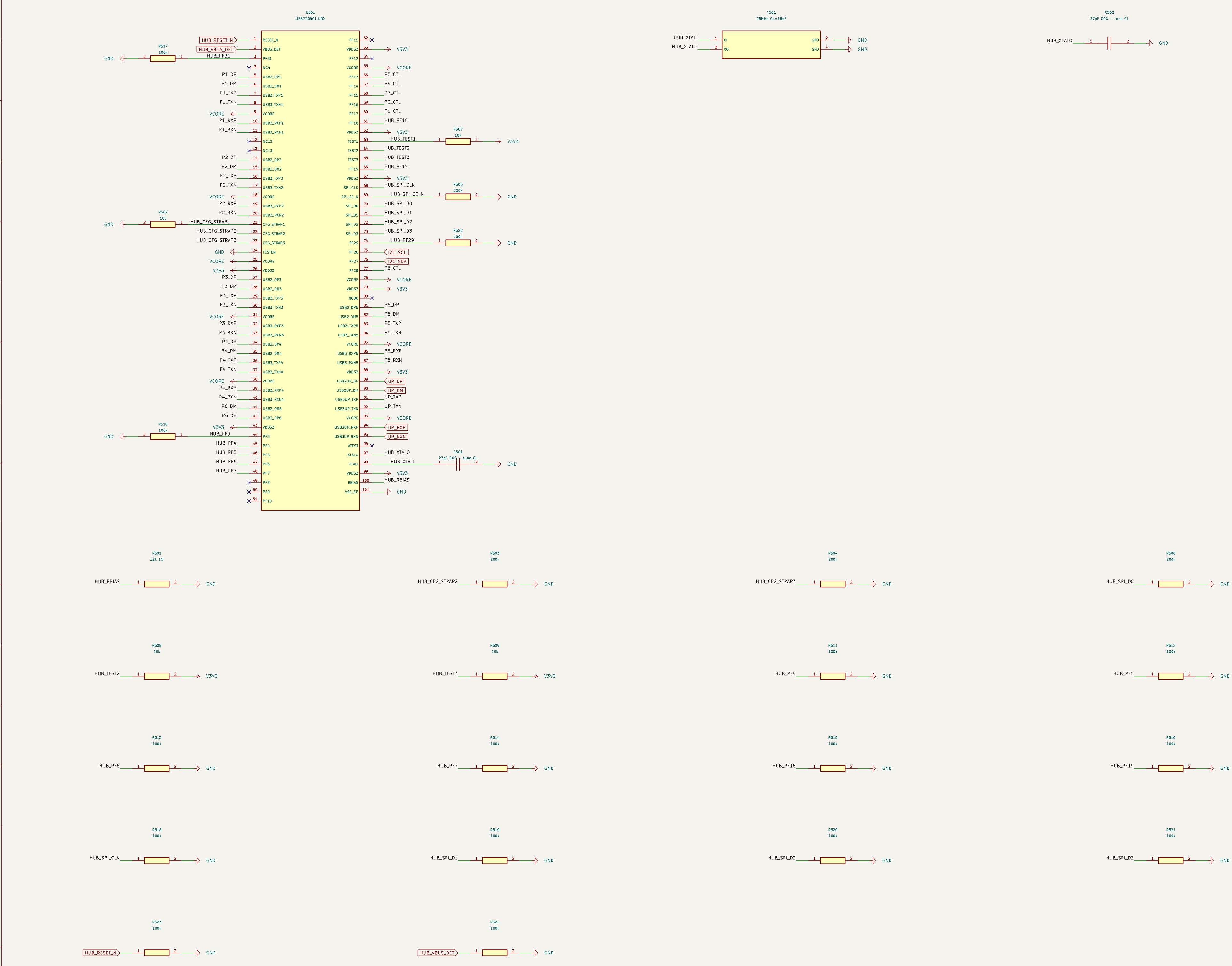
Named nets connect the circuit blocks on this sheet and across the hierarchy.

STM32G071CBT6 LQFP40: PA8(28)=UCPD1_CC1, PB15(27)=UCPD1_CC2, PB6(45)/PB7(46)=I2C1.
Source-only: UCPD1 dead-battery pins 29/32 grounded. Disable BOTH UCPD dead-battery functions before GPIO as VBAT and VREF+ tied to 3.3 V; internal VREFBUF must stay disabled. Preserve NRST and SWD; firmware required.



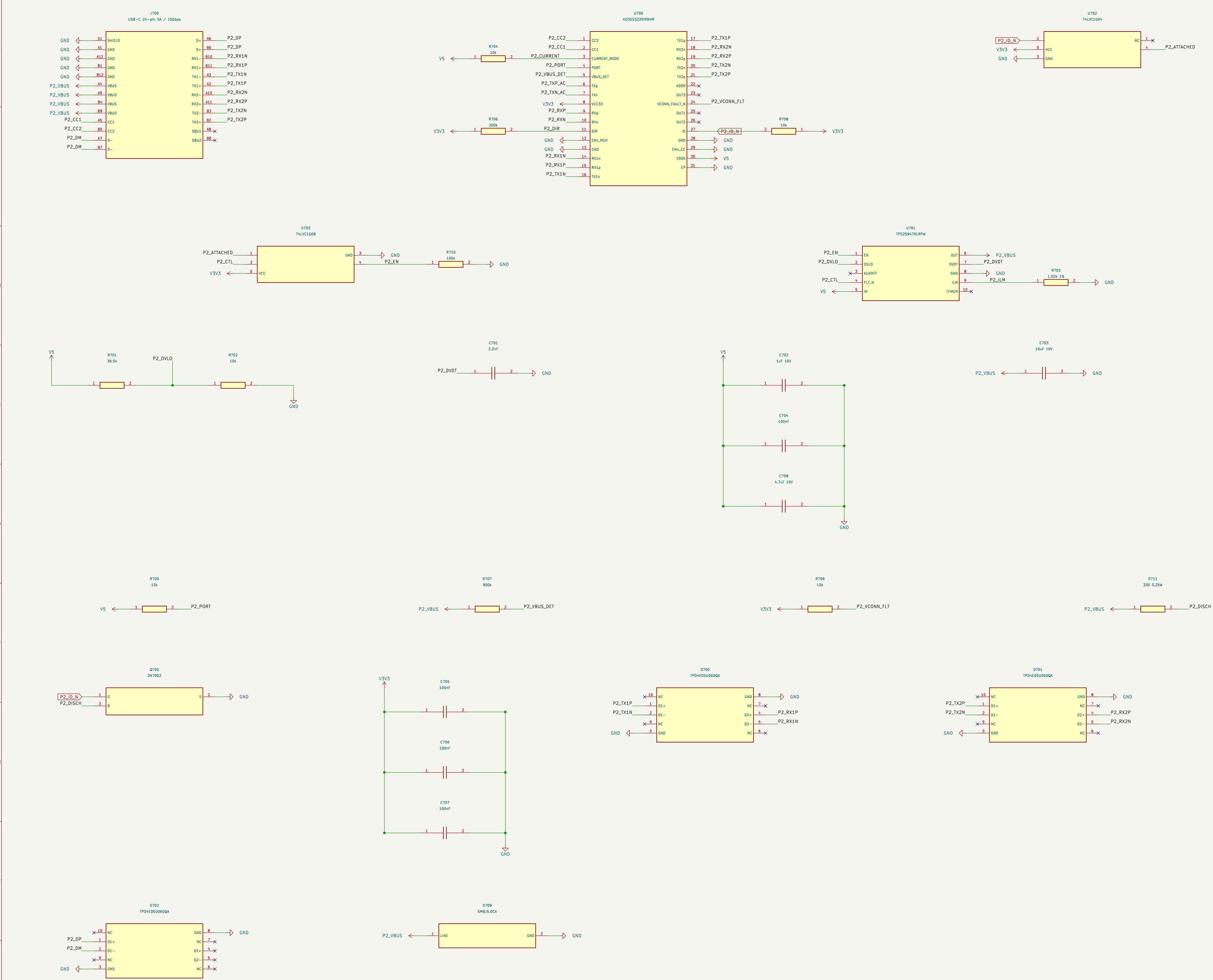
USB204C - six downstream ports

See USB 2.0 Full-Speed and High-Speed Transceiver PHYs for more details on the PHY and its signal and power coupling.



Downstream Type-C 2 - 5 V / 3 A

See USB 2.0 Full-Speed and High-Speed Transceiver PHYs for more details on the PHY and its signal and power coupling.



Downstream Type-A 2 - SuperSpeed / 900 mA

See USB 2.0 Full-Speed and High-Speed Transceiver PHYs for more details on the PHY and its signal and power coupling.

